

INDIAN ECONOMY

**LEVEL
II**



LECTURES

AN OVERVIEW ABOUT THE ZERO BUDGET
NATURAL FARMING (ZBNF)

MODULE - 71

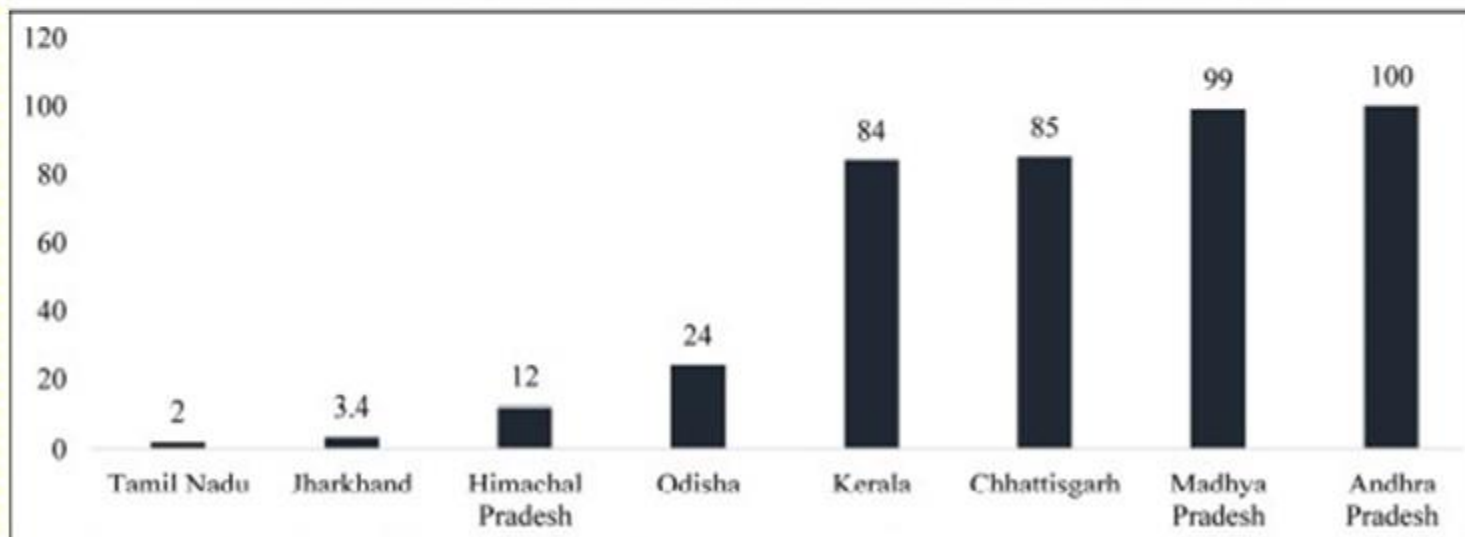


PROMOTION OF NATURAL FARMING THROUGH BHARATIYA PRAKRITIK KRISHI PADDHATI (BPKP)

- Natural farming in India is being promoted through a dedicated scheme of Bharatiya Prakritik Krishi Paddhati (BPKP).
- It promotes
 - ✓ On farm biomass recycling with major stress on biomass mulching.
 - ✓ Use of on farm cow dung and urine formulations.
 - ✓ Periodic soil aeration and exclusion of all synthetic chemical inputs.

- Under BPKP, financial assistance of Rs. 12,200 per hectare for three years is provided for cluster formation, capacity building and continuous hand holding by trained personnel, certification and residue analysis.

Figure 19: Area Covered under BPKP as on 07.12.2021 (in '000 ha).



Source: Based on data of the Ministry of Agriculture and Farmers Welfare

CONCEPT OF ZERO BUDGET NATURAL FARMING (ZBNF)



- Desi cow dung and urine
- No hybrid / GM seeds
- No fertilizers
- No pesticides

According to Subhash Palekar, 98.5% of the nutrients that plants need is obtained from air, water and sunlight, only 1.5% is from the soil.



- Subhash Palekar developed the ZBNF.
- He identified the following four aspects -
 - ✓ Seeds treated with cow dung and urine.
 - ✓ Soil rejuvenated with cow dung, cow urine and other local materials to increase microbes.
 - ✓ Cover the soil with straw and other organic matter to retain soil moisture and build humus.
 - ✓ Soil aeration for favorable soil conditions.

These are combined with natural insect management methods, when required.

ZERO BUDGET NATURAL FARMING ... AN EXAMPLE OF REGENERATIVE AGRICULTURE!



- It is an example of regenerative agriculture practice.
- It places emphasis on natural processes to re-energize the soil by increasing organic matter and help plants get natural nutrients.
- Soil is full of nutrients. Plants also need them in cooked form and these are provided by the microorganisms present in soil.
- Jeevamrut is the best source to proliferate such microbes. Dung and urine of the local indigenous breed of cow are the best ingredients to make Jeevamrut.

FOUR PILLARS OF ZBNF

JEEVAMRUTHAM

A fermented microbial culture derived primarily from cow dung and urine, jaggery, pulse flour and uncontaminated soil.

BEEJAMRUTHAM

Microbial coating of seed, seedlings by using cow dung, urine and lime.

ACCHADANA OR
MULCHING

The process of covering the top soil with cover crops and crop residues.

WHAPHASA

It means "Soil Aeration".



There is also emphasis on inter-cropping, the practice of growing many crops next to one another.

To manage insects and pests, ZBNF recommends the use of "Agniastra", "Brahmastra" and "Neemastra" and these are also based mainly on urine and dung of Indian cow breeds. The idea is that since these need not be purchased, farming remains practically "Zero Budget".

FOUR PILLARS OF ZBNF IN DETAIL

1 JIVAMRITA/JEEVAMRUTHA



- It is fermented microbial culture.
- It acts as a catalytic agent that promotes the activity of micro-organisms in the soil as well as it increases earthworm activity.
- During the 48-hour fermentation process, the aerobic and anaerobic bacteria present in the cow dung and urine multiply, as they eat-up organic ingredients like pulse flour.
- It prevents fungal and bacterial plant diseases.
- Jivamrita was needed for the first three years of the transition, after which the system becomes self-sustaining.

FOUR PILLARS OF ZBNF IN DETAIL

2 BIJAMRITA/BEEJAMRUTHA



- It is the treatment used for seeds, seedlings or any planting material.
- It is effective in protecting young roots from fungus as well as from soil-borne and seedborne diseases that commonly affect the plants after the monsoon period.
- Local cow dung is a powerful natural fungicide and cow urine is a strong anti-bacterial liquid.

FOUR PILLARS OF ZBNF IN DETAIL

3 ACCHADANA - MULCHING



- This protects topsoil during cultivation and does not destroy it by tilling. It promotes aeration and water retention in the soil.
- Straw material usually refers to the dried biomass waste of previous crops. Hence, it consists of the dead material of any living plants.
- It is also essential to develop multiple cropping patterns with monocots and dicots grown in the same field to supply all essential elements to the soil and crops.
- Monocots have one seed leaf, dicots have two seed leaves. Legumes are of the dicot group and are nitrogen-fixing plants.

FOUR PILLARS OF ZBNF IN DETAIL

4

WHAPASA - MOISTURE



- As per Palekar, plant roots does not need lot of water. According to him, what roots need is water vapour.
- Whapasa is the condition, where there are both air molecules and water molecules present in the soil and he encourages reducing irrigation, irrigating only at noon, in alternate furrows.
- Hence, it reduces the water requirement drastically!

WHY IS THE SCIENTIFIC COMMUNITY AGAINST ZBNF?

1 IT IS NOT AT ALL ZERO BUDGET!



- ZBNF is not Zero Budget.
- Many ingredients have to be purchased.
- It involves wages for hired labour, imputed value of family labour, imputed rent over owned land, costs of maintaining cows and paid-out costs on electricity and pump sets etc.

WHY IS THE SCIENTIFIC COMMUNITY AGAINST ZBNF?

2 NO RELIABLE FIELD TRIAL REPORT TILL NOW!

- There are no claims to validate higher yields. Independent studies based on field trials are not available.
- Government of Andhra Pradesh has a report, but it appears to be a self-appraisal by the implementing agency.
- One field trial is going on at G.B. Pant University of Agriculture and Technology, but its complete result is not yet available.



WHY IS THE SCIENTIFIC COMMUNITY AGAINST ZBNF?

3

ONE TREATMENT FOR VARIETY OF SOILS LACKS SCIENTIFIC BASIS!



ZBNF practitioners appear to insist on one blanket solution for all the problems of Indian soils, which clearly lacks scientific basis!

- Indian soils are poor in organic matter content. About 59% are low in available nitrogen, about 49% are low in available phosphorus, about 48% are low or medium in available potassium.
- Soils also have varying deficiencies in micronutrients. Micronutrient deficiencies also disallow the full expression of other nutrients in the soil leading to an overall decline in fertility.
- Some soils are saline or acidic due to nutrient deficiencies, certain soils are toxic.

WHY IS THE SCIENTIFIC COMMUNITY AGAINST ZBNF?

4 THE PROCESS LACKS SCIENTIFIC BASIS!



- Jeevamrut makes the nutrients available to the plants by increasing the population of soil microorganisms.
- Jeevamrut prescription is the application of 10 kg of cow dung and 10 litres of cow urine per acre per month.
- Nitrogen content of cow dung is just 0.5% and 1% in cow urine, this translates to just about 750 g of nitrogen per acre per season. This is totally inadequate.

THE GOVERNMENT SHOULD TREAD CAUTIOUSLY!



LISTEN TO THE
CONTRASTING
OPINIONS TOO!

1 In farmlands on which chemical farming has been practised intensively, the soil quality is poor and may not respond quickly to ZBNF.

2 Also, in crops such as paddy, standing water in the field inhibits soil microbial population, which in turn impacts the yield after farmers switch to ZBNF.

3 It may take few years for the effect of natural farming to be reflected in the yield.



4 The effect on the crop yields may vary, depending on the soil quality, crop and methods adopted by the farmer.

5 ICAR is yet to study the technique and its scalability in detail.

6 Hand-holding of the farmers during the initial years is inevitable.

THE GOVERNMENT SHOULD TREAD CAUTIOUSLY!

11 Farmers will be motivated, if income support is given in the initial years.

10 If the soil quality is poor or the farmer does not follow the recommended practices, crops will be susceptible to pest attacks. The Government needs to be ready with Plan B.



7 The farmers need right supply of inputs - not every farmer owns a cow, that too a native cow, as Palekar is particular about desi cow dung for Jeevamrutham.

8 The Government should also invest in building the seed banks in every village to ensure supply of desi seeds. ZBNF stresses the use of desi seeds, not hybrid / GM seeds.

9 There is every possibility of crop failure in the first year of transition to ZBNF. Pest management under ZBNF is only preventive.

LISTEN TO THE
CONTRASTING
OPINIONS TOO!



WE EXPRESS OUR
SINCERE THANKS
FOR VIEWING THIS VIDEO

Presented by



Learning Space

For Suggestions:

suggestions@learningspace.in

To Contact us:

info@learningspace.in

For Information:

info.learningspacedigital@gmail.com

Visit us at:

www.learningspacedigital.com

98499 42299